

5K The zero index and power of a power

Learning Objective	To be able to evaluate and simplify exponent forms through the use of index laws.
Today We Are	Using the zero index law and the power of a power law
Indicators of Progress	- I can use the zero index law - I can use the power of a power law

Let's Remember:

$$\begin{aligned}
 &\bullet 5^5 \times 5^3 = 5^7 \\
 &\bullet 3^2 \times 3^4 \times 3^3 = 3^{2+4+3} = 3^9 \\
 &\bullet 7^7 \div 7^4 = 7^{7-4} = 7^3
 \end{aligned}$$

What happens if I have...

$$3^3 \div 3^3 = \frac{\overset{1}{\cancel{3}} \times \overset{1}{\cancel{3}} \times \overset{1}{\cancel{3}}}{\underset{1}{\cancel{3}} \times \underset{1}{\cancel{3}} \times \underset{1}{\cancel{3}}} = 1$$

$$3^0 = ? \quad |$$

Rule for index of zero:

Any index form that has a power/index of zero = 1

Examples:

- $4^0 = 1$
- $5^0 = 1$
- $68^0 = 1$
- $458^0 = 1$
- $1000^0 = 1$



Rule for power of powers:

When raising a number in index form to a power, keep the base and multiply the indices

$$4^x = 4^6$$

$$5^x = 5^{15}$$



$$5^0 \times 6 = 1 \times 6 = 6$$

x

$$\frac{4^6}{4^4 \times 4^2} = \frac{4^6}{4^6} = 4^0 = 1$$

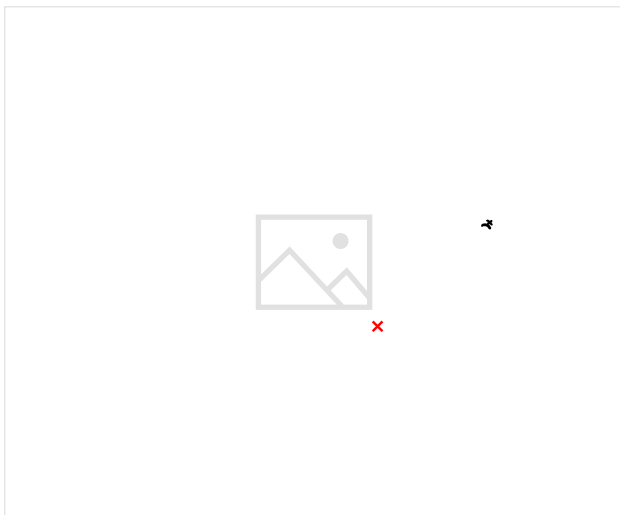
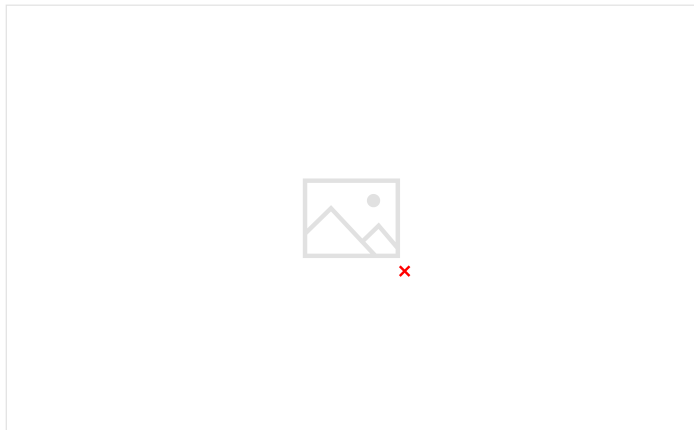
x

$$\frac{(7^6)^0 \times 7^{12}}{7^2 \times 7^6 \times 7^3} = \frac{7^0 \times 7^{12}}{7^2 \times 7^6 \times 7^3} = \frac{1 \times 7^{12}}{7^{11}} = \frac{7^{12}}{7^{11}} = 7^{12-11} = 7$$



x

x



$$\begin{aligned} (3x^5)^2 &= 3^2 x^{5 \cdot 2} \\ &= 9x^{10} \end{aligned}$$